

# Case Study Research

# Definition of Case Study

Yin defines the case study research method as

“an empirical inquiry that investigates a contemporary phenomenon within its real-life context; when the boundaries between phenomenon and context are not clearly evident; and in which multiple sources of evidence are used” (Yin, 1984, p. 23).



# Why against?

- Lack of rigor while performing research
- Concerns over generalization
- Data collection procedures are not standardized
- Case study research different from qualitative research



# Issues

Critics of the case study method often claim:

- Studying a small number of cases is insufficient for establishing reliability or generality of findings;
- The intense involvement of the researcher in the study of the case could bias the findings;
- Some critics suggest case study research is useful only as an exploratory tool or for establishing a hypothesis;
- Some would claim it is unscientific.

# Purpose of case studies

01

Explain causal links  
Describe an  
intervention

02

Illustrate certain  
topics with an  
evolution

03

Explore situations  
where inventions  
have no outputs

# When to be applied?

How or why questions are being asked

The researcher has little control over events

When the focus is contemporary phenomenon within a real-life context

# Benefits

The case study method is appropriate to:

- Answer **why**? questions;
- Gain an in depth understanding of a particular phenomenon;
- Develop an initial hypothesis;
- Establish a basis for future research;
- Develop new theories;
- Extend existing theories;
- Test existing theory.

# How many cases?

Eisenhardt (1989), suggests from 4 to 10 is ideal;

Yin (1984) and Stake (1995) suggest that one can be acceptable;

The key is to justify your choice.



# Strategies Overview

| Strategy          | Types of Questions                   | Control on events? | Contemporary events? |
|-------------------|--------------------------------------|--------------------|----------------------|
| Experiments       | How, why                             | Yes                | Yes                  |
| Survey            | Who, what, where, how many, how much | No                 | Yes                  |
| Archival Analysis | Who, what, where, how many, how much | No                 | Yes/No               |
| History           | How, why                             | No                 | No                   |
| Case Study        | How, why                             | No                 | Yes                  |

# Another Definition

The essence of a case study, the central tendency among all types of case study, is that it tries to illuminate a *decision* or a set of decisions; why they were taken; how they were implemented; and with what results (Schramm, 1971)

# Designing Case Studies

Research design is a logical plan for getting from initial set of questions to be answered to set of conclusions

Components:

- A study question
- Purpose in the case of exploratory case (propositions)
- Unit of analysis
- Logic linking between data to propositions
- Criteria for interpreting the findings

# Conditions

Construct Validity

Internal Validity

External Validity

Reliability

| Tests              | Definition                                                              | Case Study Tactic                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Construct Validity | Correct operational measure of concepts                                 | <ul style="list-style-type: none"> <li>• Use multiple sources of evidence</li> <li>• Establish chain of events</li> <li>• Have key informants review the draft case study report</li> </ul> |
| Internal Validity  | Establishing a non- spurious causal relationship (only for explanatory) | <ul style="list-style-type: none"> <li>• Do pattern matching</li> <li>• Do explanation building</li> <li>• Address rival explanation</li> <li>• Use logic models</li> </ul>                 |
| External Validity  | Establishing the domain for generalisation                              | <ul style="list-style-type: none"> <li>• Use theory in single case studies</li> <li>• Use replication in multiple case studies</li> </ul>                                                   |
| Reliability        | Repeatability of operations of the case study                           | <ul style="list-style-type: none"> <li>• Use case study protocol</li> <li>• Develop case study database</li> </ul>                                                                          |

# Theory is needed?

Requires theoretical propositions

Helps generalise from case study to theory

Analytical generalisation and not statistical generalisation

# Stages



Determine the research questions;



Decide if case study is appropriate;



Decide how many cases are appropriate;



Select the cases and determine data gathering and analysis techniques;



Prepare to collect the data;



Collect data in the field;



Evaluate and analyse the data;



Write up the research and present the findings.

\* These processes are not always linear and can be iterative

# Case Study Design

## Single Case

- Represents a critical case in testing a well formulated theory
- Represents an extreme case or unique circumstances
- Representative of a typical case
- Revelatory case – previously inaccessible to research
- Longitudinal case – how certain conditions change over time



# Case Study Design

## Multiple Case

- Replication
  - 1 - Literal replication: predicts similar results
  - 2 - Theoretical replication: predicts contrasting results for predictable reasons

When external conditions are not thought to produce much variance in the phenomena being studied, smaller number of theoretical replications is needed

Heuristics for replications:

- Settle for 2 or 3 replications when the rival theories are really different and issues at hand do not demand an excessive degree of certainty
- If rivals are subtly different and if high degree of certainty is desired, go for 5 or 6 replications

# Skills for case study research

Ask good questions

Good listener

Adaptive and  
flexible

Have clear  
understanding of  
the issues being  
studied

Be unbiased by  
preconceived  
concepts



Why the study is being done



What evidence is being researched



What variations can be anticipated



What could constitute supportive or contrary evidence for any given proposition

# Training Requirements



# Session Agenda

1. Purpose of the case studies and research questions
2. Review case study nominations and nomination procedures
3. Schedule for doing case study
  - Preparation Study
  - Arrangements for site visit
  - Conduct site visit
  - Follow-up activities
  - Preparation of case study report
  - Submission of draft report to site review



# Session Agenda

4. Review of case study protocol
  - Discussion of relevant theoretical framework and literature
  - Development or review of hypothetical logic model (if relevant)
  - In-depth discussion of protocol topics
5. Outline of case study report



# Session Agenda

## 6. Methodological Reminders

- Fieldwork procedure
- Use of evidence
- Note taking
- Other orienting topics

## 7. Reading Materials

- Sample case study reports
- Key substantive books and articles



# Case study Protocol

Protocol is a standardised agenda for the researcher's line of inquiry for a single case

Overview of the case study Project

- Background information
- Issues to be investigated
- Relevant readings

Introduction to the case study and purpose of the protocol

- Case study questions, hypothesis and propositions
- Theoretical framework for the case study



The image shows several rolled-up architectural blueprints. The blueprints are white with black lines and numbers. Some numbers visible include 2380, 5510, 810, 820, 2180, 150, 740, 970, 1385, 990, 530, 150, 2650, 5840, 4000, 150, 760, 830, 40, 1030, 1480, 890, 380, 150, 1780, 155, 425, 270, 1185, 250, 1125, 250, 2560, 2810, 2500, 3175, 3100, 1010, 5, 6, 3, 4, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100. The blueprints are arranged in a way that they appear to be part of a larger set of plans.

# Case study Protocol

- Data collection procedures
- Site to be visited
- Data collections plans
- Expected preparation before site visits
- Case study questions
- Outline of the case study report

# Case Study Questions

Level 1: questions asked for specific interviewees

Level 2: questions asked for the individual case

Level 3: questions asked of the pattern of findings across multiple cases

Level 4: questions asked for an entire study

Level 5: normative questions about policy recommendation and conclusions

|                       | From an individual                                                     | From an organisation                                                           | Study Conclusion                      |
|-----------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------|
| About an individual   | Individual behaviour<br>Individual attitudes<br>Individual perceptions | Archival records<br>Other reporters<br>behaviour,<br>attitudes,<br>perceptions | If a case study is<br>an individual   |
| About an organisation | How organisations work<br>Why organisations work                       | Personal policies<br>Organisational<br>outcomes                                | If a case study is<br>an organisation |

# Data collection sources

## Why pilot case study

- Inquiry in the case is both broad and less focused
- Help refine data collection plans

Pilot test is not a pretest

Convenience, access and geography are main criteria for selecting pilot case

Pilot case report must highlight the lessons learnt from research design and field procedures.

## Sources of data/ evidence

- Documents
- Archival records
- Interviews
- Direct observation
- Participant - observation
- Physical artifacts

## Principles of data collection

- Using multiple sources of evidence (triangulation to develop converging lines of inquiry. Strengthens construct validity)
- Creating a case study database
- Maintaining a chain of evidence

| Forms                                                                                                                                                                                                                     | Strengths                                                                                                                                                                                                                                                                                                             | Weaknesses                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Letters, memoranda, communications, agendas, announcements, minutes of meetings, written reports on events, administrative documents (proposals, progress reports, internal records), formal studies, newspaper clippings | <ul style="list-style-type: none"> <li>• Stable - can be reviewed repeatedly</li> <li>• Unobtrusive - not created as a result of the case study</li> <li>• Exact - contains exact names, references, and details of an event</li> <li>• Broad coverage - long span of time, many events, and many settings</li> </ul> | <ul style="list-style-type: none"> <li>• Retrievability- can be low</li> <li>• Biased selectively, if collection is incomplete</li> <li>• Reporting bias- reflects bias of authors</li> <li>• Access- may be deliberately blocked</li> </ul> |

## Sources of evidence (Documentation)

## Forms

Service records, organizational records, maps and charts, lists, survey data, and personal records

## Strengths

- Stable- can be reviewed repeatedly
- Unobtrusive- not created as a result of the case study
- Exact- contains exact names, references, and details of an event
- Broad coverage- long span of time, many events, and many settings
- Precise and quantitative

## Weaknesses

- Retrievability- can be low
- Biased selectively, if collection is incomplete
- Reporting bias- reflects bias of authors
- Access- may be deliberately blocked
- Accessibility due to privacy reasons

# Sources of evidence (Archival Records)

| Forms                                                                                                                                                                                                                                                                                                                                                                                                       | Strengths                                                                                                                                                          | Weaknesses                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Guided conversations, instead of structured queries. Important to</p> <ol style="list-style-type: none"> <li>1) follow your own line of inquiry, as reflected by the case protocol;</li> <li>2) to ask actual conversational questions in an unbiased manner.</li> <li>3) Open ended questions (espousing facts and opinions). Could even be a focused, short interview.</li> <li>4) Surveys.</li> </ol> | <ul style="list-style-type: none"> <li>• Targeted- focused directly on the case study topic</li> <li>• Insightful- provides perceived causal inferences</li> </ul> | <ul style="list-style-type: none"> <li>• Bias due to poorly constructed questions</li> <li>• Response bias</li> <li>• Inaccuracies due to poor recall</li> <li>• Reflexivity- interviewee gives what interviewer wants to hear</li> </ul> |

# Sources of evidence (Interviews)



| Forms                                                                                                                                                                           | Strengths                                                                                                                                           | Weaknesses                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Behavior or environmental conditions worth observing at a site.</p> <p>Ranges from formal to casual data collection activities.</p> <p>Have more than a single observer.</p> | <ul style="list-style-type: none"> <li>• Reality- covers information in the real time</li> <li>• Contextual- covers context of the event</li> </ul> | <ul style="list-style-type: none"> <li>• Time consuming</li> <li>• Selectivity- unless broad coverage</li> <li>• Reflexivity- event may proceed differently because it is being observed</li> <li>• Cost- hours needed by human observation</li> </ul> |

## Sources of evidence (Direct Observations)

## Forms

You may assume a variety of roles within a case study situation and may actually participate in the events being studied.

Usually used in anthropological studies

## Strengths

- Reality - covers information in the real time
- Contextual - covers context of the event
- Insightful into personal behaviors and motives

## Weaknesses

- Time consuming
- Selectivity - unless broad coverage
- Reflexivity - event may proceed differently because it is being observed
- Cost - hours needed by human observation
- Bias due to investigators' manipulation of events

# Sources of evidence (Participant observation)

| Forms                                                                                        | Strengths                                                                                                                             | Weaknesses                                                                              |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| A technological device, a tool or instrument, a work of art, or some other physical evidence | <ul style="list-style-type: none"> <li>• Insightful into cultural features</li> <li>• Insightful into technical operations</li> </ul> | <ul style="list-style-type: none"> <li>• Selectivity</li> <li>• Availability</li> </ul> |

## Sources of evidence (Physical Artifacts)

# Principles of data collection

## 1. Use multiple evidence

- Triangulation types (Patton, 1987)
  - Of data sources (data triangulation)
  - Among different evaluators (investigators triangulation)
  - Of perspectives to the same data set (theory triangulation)
  - Of methods (methodological triangulation)

## 2. Create a case study database: Organizing and documenting the data collected

- Two collections
  - The data or evidence base
  - The report of the researcher, whether in paper, report or book format
- Uses notes, documents, tabular material, and narratives

## 3. Maintain a chain of evidence

- Increases reliability
- External observer should be able to trace the steps in either direction
- Report should make sufficient citations to relevant portions
- Revel actual evidence and circumstance 
- Consistency with protocol

## Steps of analysis

- Examining
- Categorizing
- Tabulating
- Creating a data display
- Testing
- Combining qualitative and quantitative evidence to address initial propositions

## Techniques for analysis

- Pattern matching
- Explanation building
- Time-series analysis
- Logic model
- Cross-case synthesis

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A good  
analysis  
should

Attend to all the evidence  
(including the rival hypothesis)

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Must address all major rival  
interpretations

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Address most significant part  
of your case study

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Use your own prior expert  
knowledge

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# Specific Analytical techniques

## Pattern matching

- Compares an empirically based pattern with a predicted one to strengthen internal validity
- Rival explanations as patterns

## Explanation building

- Relevant to explanatory case studies
- Iterative nature of explanation building
- Risk of drifting away from the original topic of interest

# Specific Analytical techniques

## Time-series analysis

- Only a single dependent and independent variable
- Detailed and precise tracing of events
- Trend matching with stated propositions, rival trends, or trends based upon artifacts
- Chronologies to investigate presumed causal events
- Interruption in time series as a potential causal relationship

## Logic models

- Stipulates a complex chain of events over time
- Staged in repeated cause- effect relationship
- Analysis can also entertain rival chains of events, and spurious external events
- Could be individual level or organizational level logic model



# Reporting Case Studies

## Formats of case study report

- Classic single narrative, suitable for a book and not journal
- Multiple-case version with narratives, including cross- case analysis and results
- Presented in short question- answer formats, without narratives
- Entire report dedicated to cross- case analysis, descriptive or explanatory

# Procedure in doing a case study report

## When and how to start composing

- Bibliography and methodology section must be started
- Followed by descriptive data about the case being studied

## Case identities: real or anonymous

- Full disclosure is the most desirable option, helping reader link in previous research, and helps ease of review

## The review of the draft case study: The validating procedure

- Draft reviewed by peers, informants, and participants of the case

# Some classic texts about case

Eisenhardt, K. M. (1989). Building theories from case study research. *Academy of Management Review*, 14(4), 352-550.

Simons, H. (2009). *Case study research in practice*, London: Sage.

Stake, R. E. (1995). *The art of case study research*. Thousand Oaks, CA: Sage Publications Ltd.

Yin, R. K. (1984). *Case study research: Design and methods*. Newbury Park, CA: Sage.

# References

[https://unvi.edu.ba/Files/DLInt/sample/Action%20research/DBA502 Slides 3 Case study decisions and techniques v2.ppt](https://unvi.edu.ba/Files/DLInt/sample/Action%20research/DBA502_Slides_3_Case_study_decisions_and_techniques_v2.ppt)

<https://pt.slideshare.net/pavan7soni/case-study-research-by-robert-yin-2003>